

PROJECT PROPOSAL

Project Initialization and Planning Phase

Date	20 FEB 2026
Team ID	LTVIP2026TMIDS5743
Project Title	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	3 Marks

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Project Overview	
Objective	The primary objective of this project is to analyze and visualize housing market trends by examining sale prices and key property features using Tableau. The project aims to provide insights into market patterns, price fluctuations, and factors influencing home values, helping stakeholders make data-driven decisions.
Scope	• Geographical Coverage: Focus on a specific city, region, or nationwide data. • Data Sources: Utilize publicly available datasets (e.g., , Kaggle, government housing data). • Analysis Areas: Price trends over time, location-based variations, property features affecting prices. • Visualization Methods: Interactive dashboards, Overall data, Pie chart, Bar charts, and Histogram.
Problem Statement	
Description	In this project for ABC company, the goal is to address challenges in understanding the factors that influence house prices and sales trends. By analyzing comprehensive housing data, including the total sales by years since renovation, house age distribution by the number of bathrooms, bedrooms, and floors, and the impact of renovations on house age, the company aims to uncover key insights. Utilizing TABLEAU for this analysis, the objective is to visualize and interpret patterns in the housing market to inform strategic decisions, optimize pricing strategies, and enhance overall market competitiveness. Key stakeholders include real estate analysts, marketing teams, and company executives, all of whom will benefit from a deeper understanding of housing market dynamics
Impact	• Helps potential buyers and sellers make informed decisions. • Assists real estate professionals in understanding pricing trends. • Provides policymakers with insights into housing affordability and market fluctuations.

Proposed Solution	
Approach	Collect and preprocess historical housing market data. Use Tableau to create interactive visualizations. Apply data analysis techniques to identify patterns and correlations. Develop dashboards for easy interpretation of housing trends.
Key Features	Dynamic Visualizations: Interactive dashboards showing trends over time. Geospatial Analysis: Heat maps highlighting price variations by location. Comparative Analysis: Charts comparing features like square footage vs. price. Predictive Insights: Basic forecasting of price trends using historical data.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (intel i7)
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Data Visualization Frameworks	Tableau public
Libraries	Additional libraries	Flask Package
Development Environment	Uploading project	GitHub, Drive
Data		
Data	Source, size, format	Kaggle Dataset file (Transformed Housing Data 2), CSV 2.41 MB